Course Information

STAT 51100 Summer 2026
Statistical Methods
(Version of May 28 2026)

Instructor: Evidence Simbarashe Matangi
Office: MATH 246
Office Hours:

Monday 10:30 am – 11:30 am by Evidence Matangi
Tuesday and Friday 5:30 – 6:30 pm by Hongjie Liu
Monday and Tuesday 1pm – 2 pm by Nianyao Du
Thursday and Friday 8 pm – 9 pm by Shishang Wu
Tuesday and Wednesday 11 am – 12 pm by Haoyu Hu
Wednesday and Thursday 4 – 5 pm by Minhyeok Kim

All times are based on Eastern Daily Saving Time

Instructor Office Hours Link: Zoom

Meeting link
https://purdue.webex.com/meet/ematangi
Meeting ID: 647 480 138; Access code 647 480 138

Email: ematangi@purdue.edu Make sure you will have [**STAT511**] in your email header.

TA's email and weblinks for office hours:

TA Name	Email	WebEx room link
Haoyu Hu	hu943@purdue.edu	https://purdue-student.webex.com/meet/hu943
Minhyeok Kim	kim4660@purdue.edu	https://purdue-student.webex.com/meet/kim4660
Hongjie Liu	liu4114@purdue.edu	https://purdue.webex.com/meet/liu4114
Shishang Wu	wu1396@purdue.edu	https://purdue-student.webex.com/meet/wu1396
Nianyao Du	du205@purdue.edu	https://purdue.webex.com/meet/du205

Email the instructor first if you have any grading issues. You are more than welcome to email TA for any other questions based on your last name.

Textbookand Cengage WebAssign: Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Ed., Required. (E- copy of this textbook will be available in the WebAssign system). We will be covering chapters 1 to 9. The instructions to access Webassign, are in the WebAssign module on Brightspace. The cost to access these is an out-of-pocket cost of **$112** for one course (payment will be direct to Cengage WebAssign) and in the event that one has more than one course, there is an option for unlimited access at **$170** for 1 semester.

Web Page: The course website is in Brightspace. Brightspace contains information relevant to the course. Such information includes lecture notes, homework assignments, reading assignments and question sheets, dates of exams, midterm and final review sheets, and changes to office hours. Please check Brightspace

regularly for updates.

Lectures: Lecture notes will be available on Brightspace, along with lecture videos. (*These were developed by Prof. Lingsong Zhang, for departmental resources and quality control.*)

TA/Grader: All homework will be automatically graded in the WebAssign system. If you have questions about the grades, please ask the instructor directly. TA will hold office hours. The time and location are announced above.

Final Grade: Your final grade will depend on the following components with these proportions: homework (35%), midterm (30%), final exam (35%). The proportions may be adjusted by up to 5% in either direction at the instructor's discretion but will be the same for every student in the class. Any adjustments to these proportions will be announced after the second midterm and before the end of classes. The percentage grades needed to achieve an A, B, C, or D will follow approximately the following scale: 90 – 100 = A range, 80 – 89 = B range, 70 – 79 = C range, 55 – 69 = D range, 0 – 54 = F range. The instructor then will assign "- ", "+" within each range.

Unless there is a calculation error, final grades will not be changed. With that being said, "This is my last semester", "I won't graduate if I get so and so grade", "I will lose my scholarship", "I am on probation", or "I need a certain grade to maintain my financial assistance" ARE NOT LEGITIMATE REASONS TO BUMP UP FINAL GRADES.

Software: After discussing various subjects, I will usually show how to use R to do automatic computations and some visualization. Using a computer allows us to work with larger data sets than would be possible by hand, and to experiment with the data and view the data graphically. However, I consider it important for a student to first solve typical examples by hand and calculator (to better understand the procedures), and only then perhaps use the computer.

Reading Assignments: There will be required reading for every topic: the sections to be covered in this chapter. It helps when you read about a topic *before* it is covered in class (*herein watching the video*). This does not mean that I expect you to learn it on your own. Rather, your reading before the class should be a "first pass" at the subject. The first time through, I just want you to read through it quickly, to get a general idea of the material – the "big picture". Don't get bogged down in formulas or details; just try to get a rough idea of the material and get familiar with the vocabulary. This will prepare you for what is to come in the class and will make the class easier to follow. If, as you are reading, you find something hard to understand, don't be alarmed or discouraged. Just make a note of any parts you find confusing, or any questions that occur to you as you read. Often, you will find that those questions are cleared up in the following class. If not, please ask! Later, as you are working on problem sets and studying for tests, you will find it helpful to read the material again. This time, read at a much more detailed level. It will be a lot easier to follow then, since you have already covered the material in class.

Examinations: There will be one midterm examination and one final examination, online through Brightspace. These will contribute to most of your final grade.

Midterm is due on July 8, 2026, 11:59 pm.
Midterm window (starting time): July 6, 2026, 12:00 am to July 9, 2026, 9:59 am
Midterm duration: 2.5 hours

The midterm will cover up to chapters 1 – 4.2 and the final examination will cover 4.2 to chapter 9. Each examination will have both mathematical and conceptual (written) components.

Final exam is due on August 6, 2026, 11:59 pm.
Final exam window (starting time): Aug 3, 2026, 12:00 am to August 7, 2026, 9:59 am

Final duration: 2.5 hours

Exam Rules: There will be NO MAKE-UP MIDTERM or FINAL EXAM unless there is an emergency in which case you are required to provide proper documentation, please plan ahead of time. The following situations are the only legitimate reasons to take the make-up exam: sickness (need an official letter from the doctor), death in the family, military service call, sports game (need letter from the coach). However, having a job interview, attending a conference, workshop, or contest, traveling home, and cheap airfare won't make you eligible to take the makeup exam. No exceptions to this policy.

Smart devices Cellphones, iPad, Laptops, or any devices capable of wireless communications are not to be used as calculators during exams. Any other calculators are allowed for calculations in the exams.

The exams will be performed in the Brightspace using a lock-down browser with webcam for exam integrity. Inconsistent submission will result in 0 points for the entire exam.

The exams are open books and open notes. However, materials in smart devices and computers are not allowed. Please print the needed probability tables for your usage ahead of time.

Homework: Homework will be due in WebAssign mostly on the next business day (11:59pm) after one chapter is finished. The homework will be assigned for each chapter (some chapters may have two assignments), and the due date will be posted in advance. Please try to work on problems as early as possible. Late homework is not allowed in this course. In extenuating circumstances due to family emergencies, we will allow for reasonable extensions, as we have a time crunch on the course. Time management is a skill to champion one's good success in this course.
Homework performs three vital functions in this course:

i. it gives you an opportunity to practice what you have learned and to understand concepts by using them;
ii. it gives you feedback on what you understand and on what areas need more work;
iii. it contributes to your final grade.

Of these, (iii) is the *least* important, since most of your grade is determined by examinations. You are encouraged to use homework as a learning tool. The homework assignments are quite lengthy, and I recommend that you get an early start on them and solve a few problems each day. You may also wish to discuss homework with your classmates. Group discussions and study sessions can be a useful tool for learning. However, outright copying is unacceptable, as well as pointless, and will be penalized. A good rule of thumb is that it is fine for you to talk with others about how to solve a problem, but then go and write it up yourself, possibly comparing answers afterwards if you are unsure. Remember that if you copy from a classmate without understanding it, only your classmate will pass the exam. If blatant copying is detected, all parties involved (copier and copied) will receive a score of zero for that assignment. This type of behavior will also be reported to the school.

Your WebAssign homework scores will be automatically saved in Brightspace. Please make sure the scores are consistent across these two platforms. Score inconsistency is hard to adjust after the final grade is assigned.

Grades question: If you find any errors in terms of homework grading, please check with the instructor first. The Instructor usually will explain why the points are off or help you to check with the grader. You may get some points back since errors in grading may exist.

Academic Integrity: We take academic integrity very seriously in this course. The only true way to get an education is through hard work and striving to understand concepts on your own. The penalty for academic misconduct on any assignment, exam, or final project is the failure of the course with a referral to the Dean of Students for further sanctions. Cheating on the assignments, midterm, or final exam results in an "F" for

the course. Note that we punish not only the person who cheats but also the person who enables the cheater. When it comes to academic misconduct we have zero tolerance. On this note, last several years I caught a ring of 7 cheaters on the final exam, they all received an "F" for the course.

Official Language: While speaking another language(s) is an advantage, I expect you to communicate in English with your fellow classmates in class and during office hours. English is the only spoken language in the classroom.

General Comments: Although this is not a math course, there is a fair amount of calculus discussed in the course, especially integration. I expect you to know how to integrate basic functions. As in many other science courses, we will make use of mathematics quite extensively, and most questions will have some quantitative component.

A good calculator that has some statistical functions and is easy to use will be a definite asset. (You may want to check http://www.geocities.com/calculatorhelp/ for help with your calculator.) However, the primary focus of this course is on learning how to do a good science. Doing science well requires, among other things, a good experimental design and a correct and appropriate statistical analysis of the scientific data. Therefore, knowing when and when not to use a certain statistical method, and why, and how to interpret the results, are all at least as important as knowing how to carry out the calculations. To do well in this course, you must be prepared to master all these areas.

My Expectations: I expect that you will work hard in this course. I expect you to come to each class (*herein watching each video*) prepared to listen and understand. I expect that you will ask questions if things are not clear. I expect that you will use the textbook and other resources and will read material as assigned. I expect you to attend class regularly, and that you will promptly catch up on any classes you miss. I expect you to make an honest attempt at assigned homework, and to ask for help when you need it. I expect you to behave appropriately and politely towards me and your fellow classmates at all times. This includes remaining quiet when others are speaking and being patient with the questions of others. I expect you to not misrepresent the work of others as your own, and to neither give nor receive unauthorized aid in examinations.

Your Expectations: You can expect that I will do my best to explain the material in a way that makes sense to you. Sometimes I will need help and feedback from you to figure out the best way to illustrate something. I will listen to your questions with respect and never ridicule; if the answer to your question is beyond the scope of this course, I will discuss it with you outside of class. I will give you fair notice of all assignments and tests and do my best to let you know what is required of you. I will attempt to evaluate your work fairly and assign grades appropriate to your performance. If you have other suggestions, please let me know. I will do my best to make this course a success for all of us.

EMERGENCY PREPAREDNESS□

EMERGENCY NOTIFICATION PROCEDURES are based on a simple concept – if you hear a fire alarm inside, proceed outside. If you hear a siren outside, proceed inside.

- Indoor Fire Alarms mean to stop class or research and immediately evacuate the building. o Proceed to your Emergency Assembly Area away from building doors. Remain outside until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

- All Hazards Outdoor Emergency Warning Sirens mean to immediately seek shelter (Shelter in Place) in a safe location within the closest building. o "Shelter in place" means seeking immediate shelter inside a building or University residence. This course of action may need to be taken during a tornado, a civil disturbance including a shooting or release of hazardous materials in the outside air. Once safely inside, find out more details about the emergency*. Remain in place until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

**In both cases, you should seek additional clarifying information by all means possible…Purdue Home page, email alert, TV, radio, etc…review the Purdue Emergency Warning Notification System multi-communication layers at http://www.purdue.edu/ehps/emergency_preparedness/warning-system.html*

EMERGENCY RESPONSE PROCEDURES:□

- Review the Emergency Procedures Guidelines
  https://www.purdue.edu/emergency_preparedness/flipchart/index.html

- Review the Building Emergency Plan (available from the building deputy) for:
  - evacuation routes, exit points, and emergency assembly area
  - when and how to evacuate the building.
  - shelter in place procedures and locations
  - additional building specific procedures and requirements.

EMERGENCY PREPAREDNESS AWARENESS VIDEOS□

- "Shots Fired on Campus: When Lightning Strikes," is a 20-minute active shooter awareness video that illustrates what to look for and how to prepare and react to this type of incident. See: http://www.purdue.edu/securePurdue/news/2010/emergency-preparedness-shots-fired-on-campus-video.cfm
  (Link is also located on the EP website)

MORE INFORMATION□

Reference the Emergency Preparedness web site for additional information:7
http://www.purdue.edu/emergency_preparedness